TP1 VTREF (+3V3_CORE)
TP2 GND
TP3 SWDIO
TP4 SWCLK
TP5 RUN
+3V3_CORE
GND
SWDIO
SWCLK
RUN

NORTH / J1.1-8 = N1-N8
J1
RUN
SWDIO
SWCLK
BOOTSEL_REQ
GPIO24
GPIO25
USB_DM
USB_DP
N 1-8

WEST / J4.1-8 = W1-W8
J4
CV_W1_GPIO8
CV_W2_GPIO9
CV_W3_GPIO5
CV_W4_GPIO4
CV_W5_GPIO2
CV_W6_GPIO0
CV_W7_GPIO3
CV_W8_GPIO1
W 1-8

SOUTH / J3.1-8 = S1-S8
J3
CV_S1_GPIO6
CV_S2_GPIO7
CV_S3_GPIO15
CV_S4_GPIO14
CV_S5_GPIO13
CV_S6_GPIO12
CV_S7_GPIO11
CV_S8_GPIO10
S 1-8

EAST / J2.1-8 = E1-E8
J2
CV_E1_GPIO29_ADC3
CV_E2_GPIO28_ADC2
CV_E3_GPIO27_ADC1
CV_E4_GPIO26_ADC0
CV_E5_GPIO19
CV_E6_GPIO18
CV_E7_GPIO17
CV_E8_GPIO16
E 1-8

KEY / POWER
J5 K-NW
J6 K-NE
J7 K-SW
GND
+3V3_IN

R1
+3V3_IN
+3V3_CORE
0R / FB INPUT FILTER
R1 default: 0 ohm.
Replace with same-footprint ferrite bead only after noise / ADC / EMI evaluation.

OPTIONAL EXTERNAL FLASH ASSEMBLY
RP2354A: U2, R7, R8, C17 = DNP (internal 2MB flash)
RP2350A: U2, R7, R8, C17 = FIT (external 16MB flash)
U2
FLASH_CS_EXT
QSPI_SD1
QSPI_SD2
CS#
DO
IO2
GND
VCC
IO3
CLK
DI
+3V3_CORE
QSPI_SD3
QSPI_SCLK
QSPI_SD0
W25Q128JVSIQ (RP2350A ONLY / DNP FOR RP2354A)
QSPI_SS
FLASH_CS_EXT
0R (RP2350A ONLY / DNP RP2354A)
R8
10k (RP2350A ONLY / DNP RP2354A)
C17
100n (RP2350A ONLY / DNP RP2354A)

U1
ADC_AVDD
DVDD
IOVDD
QSPI_IOVDD
USB_OTP_VDD
VREG_AVDD
VREG_PGND
EP
RUN
SWCLK
SWDIO
USB_DM_MCU
USB_DP_MCU
USB_DM
USB_DP
XIN
XOUT
+1V1
VREG_FB
VREG_LX
VREG_VIN
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SS
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
CV_W6_GPIO0
CV_W8_GPIO1
CV_W5_GPIO2
CV_W7_GPIO3
CV_W4_GPIO4
CV_W3_GPIO5
CV_S1_GPIO6
CV_S2_GPIO7
CV_W1_GPIO8
CV_W2_GPIO9
CV_S8_GPIO10
CV_S7_GPIO11
CV_S6_GPIO12
CV_S5_GPIO13
CV_S4_GPIO14
CV_S3_GPIO15
CV_E8_GPIO16
CV_E7_GPIO17
CV_E6_GPIO18
CV_E5_GPIO19
CV_E4_GPIO26_ADC0
CV_E3_GPIO27_ADC1
CV_E2_GPIO28_ADC2
CV_E1_GPIO29_ADC3
RP2354A

J8
AUX GPIO20-23 / 2.54mm PTH

POWER: C11 VREG_VIN, C12 +1V1, C13 VREG_AVDD, C14 bulk input
C1 100n
C2 100n
C3 100n
C4 100n
C5 100n
C6 100n
C7 100n
C8 100n
C9 100n
C10 100n
C11 4.7u
C12 4.7u
C13 4.7u
C14 10u

BOOTSEL: N4 -> R4 1k -> QSPI_SS
QSPI_SS
R4 1k
BOOTSEL_REQ
USB_DP_MCU
R5 27
USB_DP
USB_DM_MCU
R6 27
USB_DM
+3V3_CORE
R2 33
VREG_AVDD

INTERNAL REGULATOR
L1
VREG_LX
+1V1
AOTA-B201610S3R3-101-T

12 MHz CLOCK / USB SERIES
XIN
XOUT_XTAL
C15 15p
C16 15p
X1
12MHz
R3 1k
XOUT

Schematic	Schematic1	Create at	2026-10-27
		Update at	2026-11-04
Board	Board1	Page	1.Core-S-RP235xA
Drawn		CraftVia-Core-S-RP235xA-v1-Rev0	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	CraftVia.org

R7